



Computer
Science

CSC380: Principles of Data Science

Probability 1

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- What is probability?
- Events
- Calculating probabilities

What is probability?

What is probability?

- Suppose I flip a coin, What is the probability it will come up heads? Most people say 50%, but why?
- “Nolan’s new movie is coming out next weekend! There’s a 100% chance you’re going to love it.”



Interpreting probabilities

Basically two different ways to interpret:

- Objective probability
 - based on logical analysis or long-run frequency of an event. It's derived from known facts, symmetry, or repeated experiments.
- Subjective probability
 - based on personal belief, opinion, or information about how likely an event is, especially when there's uncertainty or limited data.

Objective or Subjective?

←  r/AskReddit • 2 yr. ago

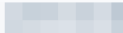
What statistically improbable thing happened to you?



 • 2y ago

When I was a teenager I picked up a hitchhiker and then a few years later the same guy picked me up when I was walking after I ran out of gas. Never saw him before or after those two occasions.



 • 2y ago

Got attacked by a robin in the morning, then attacked by a hawk 3 hours later. Weird day.



18K



Award



Share



It is a subjective probability: a belief based on their perception of how rare or meaningful the coincidence is, not a calculation based on statistical data.

Subjective probability

- Probabilities aren't in the world itself; they're in our knowledge/beliefs about the world.
- Can assign a probability to the truth of any statement that I have a degree of belief about.

We will focus on **objective probability** in this class.

Objective probability

- The probability of an event represents the long run proportion of the time the event occurs under repeated, controlled experimentation.
 - e.g. 00011101001111101000110
- Famous experiments in history on coin tosses

Experimenter	# Tosses	# Heads	Half # Tosses
De Morgan	4092	2048	2046
Buffon	4040	2048	2020
Feller	10000	4979	5000
Pearson	24000	12012	12000

Outcome, Event and Probability

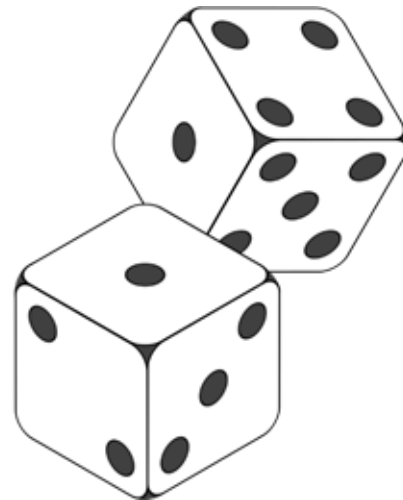
Random Events and Probability

Suppose we roll two fair dice...



Suppose we roll two fair dice...

- ◆ What are the possible outcomes?
- ◆ What is the *probability* of the following:
 - ◆ rolling **even** numbers?
 - ◆ having two numbers sum to 6?
 - ◆ If one die rolls 1, the second die also rolling 1?



...this is a random process.

How to formalize all these quantitatively?

Outcome

- Outcome is a **single** result/observation of a **random experiment**.
- Experiment 1: You flip a coin once.
 - The outcomes are: "Heads" or "Tails"
- Experiment 2: You roll a 6-sided die.
 - The outcomes are: 1, or 2, or 3, or 4, or 5, or 6
- Experiment 3: You tap "shuffle play" on your favorite Spotify playlist with 100 songs and the first song played.
 - The outcomes are: the 1st song in the list, or the 2nd song,

The Sample Space

- The set of **all** possible outcomes of a random experiment is called the sample space, written as S .
- In math, the standard notation for a set is to write the individual members in curly braces:
 - $S = \{\text{Outcome1}, \text{Outcome2}, \dots, \}$
- Useful to visualize the sample space with an actual space.

The Sample Space

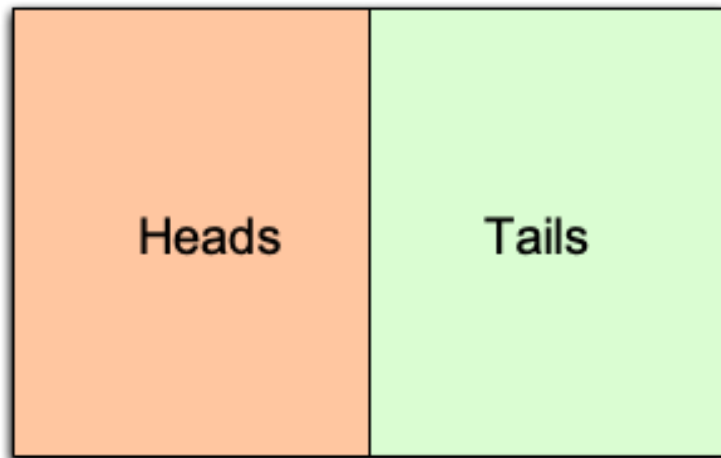


Figure: Visualization of a Sample Space

Examples of Sample Spaces

What's the sample space for a single coin flip?

- $S = \{\text{Heads}, \text{Tails}\}$



Examples of Sample Spaces

What is the sample space of rolling a die?

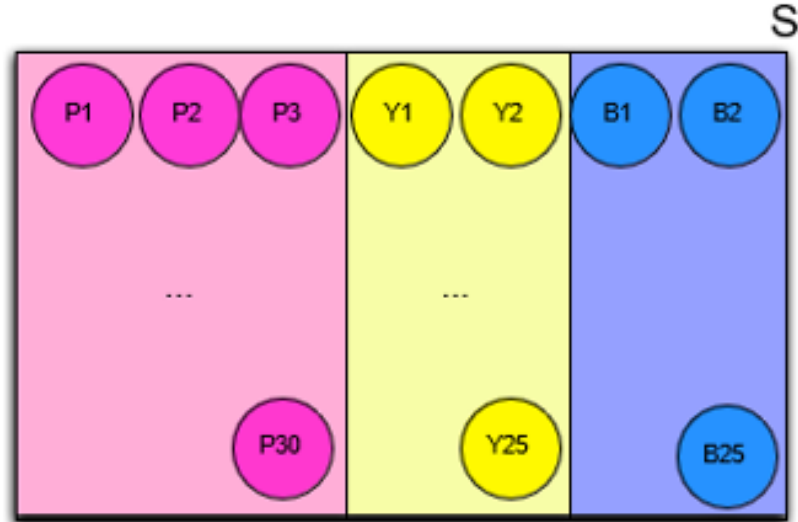
- $S = \{1, 2, 3, 4, 5, 6\}$

1	2	3
4	5	6

Examples of Sample Spaces

What is the sample space of drawing a ball out of a box containing 30 pink, 25 yellow, and 25 blue balls?

• $S = \{P1, P2, \dots, P30, Y1, \dots, Y25, B1, \dots, B25\}$



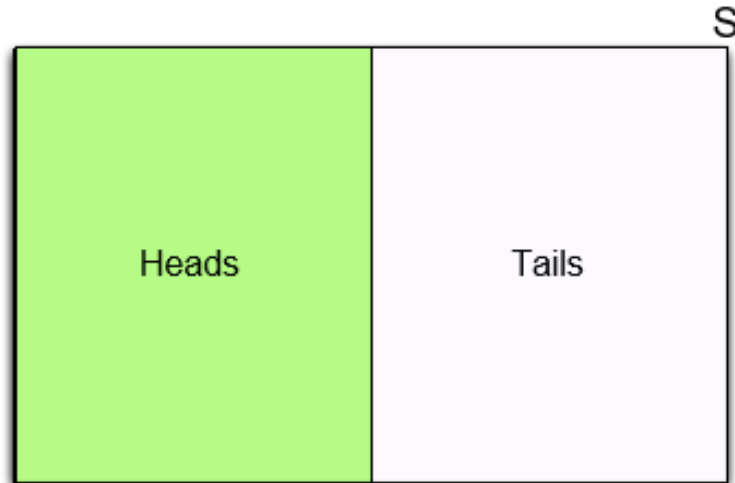
Events

- An event E is a **subset** of the sample space.
- An event E is a **set** of outcomes
 - When we make a particular observation, it is either “in” E or not.
 - Helpful to think about events as propositions (TRUE/FALSE): TRUE when the outcome is among the elements of the event set, and FALSE otherwise.
 - Is 4 in event $E = \{2, 4, 6\}$?  YES $\rightarrow$ the **proposition is TRUE**
 - Is 4 in event $F = \{1, 3, 5\}$?  NO $\rightarrow$ the **proposition is FALSE**

Examples of Events

What's the event set corresponding to the following propositions?

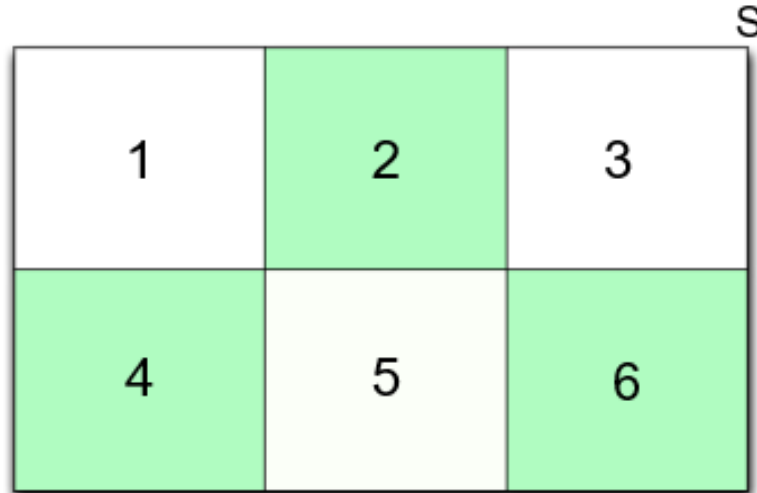
- “The coin comes up heads”
- $E = \{\text{Heads}\}$



Examples of Events

What's the event set corresponding to the following propositions?

- “The die comes up an even number”
- $E = \{2, 4, 6\}$



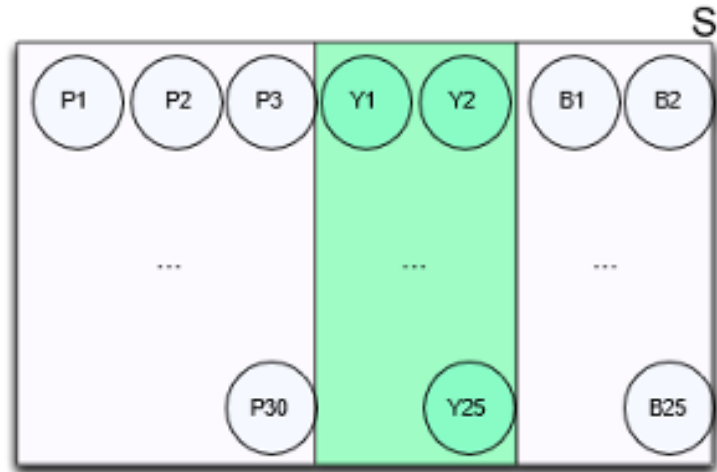
A 2x3 grid representing a sample space S for a die roll. The cells contain numbers 1 through 6. Cells with even numbers (2, 4, 6) are shaded green, representing the event E .

1	2	3
4	5	6

Examples of Events

What's the event set corresponding to the following propositions?

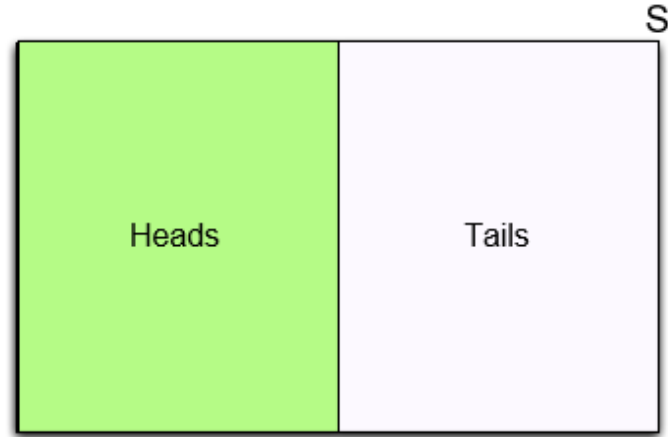
- “A yellow ball is chosen”
- $E = \{Y1, Y2, \dots, Y25\}$



Calculating Probabilities

Calculating probability

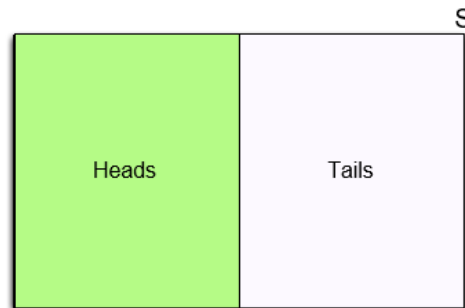
- We can think of the probability of an event E as its area, where S (sample space) always has a total area of 1.0
- So, the probability of E is the fraction of S that it takes up.



Calculating probability using symmetry

- If we have a sample space for which every outcome is equally likely (has the same area), then we can find event probabilities easily.
- Classic probability model to get the probability of an event E :

$$P(E) = \frac{\text{\#outcomes in } E}{\text{\#outcomes in } S}$$



Probability as Area

What is the probability of

$$P(E) = \frac{\text{\#outcomes in } E}{\text{\#outcomes in } S}$$

- Rolling a fair die and see an even number?

- $E = \{2, 4, 6\}$

- $$P(E) = \frac{\text{\#}\{2, 4, 6\}}{\text{\#}\{1, 2, 3, 4, 5, 6\}} = \frac{3}{6} = \frac{1}{2}$$

1	2 $P(2) = 1/6$	3
4 $P(4) = 1/6$	5	6 $P(6) = 1/6$

$$P(S) = 1$$

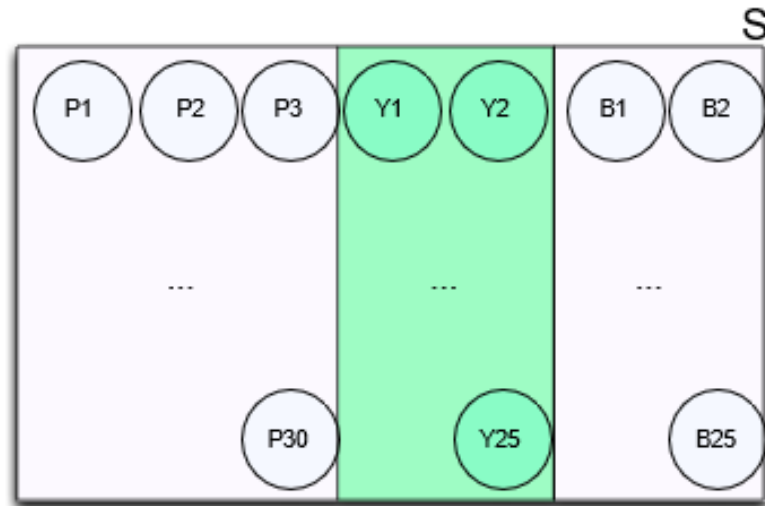
$$P(\text{Even}) = P(2) + P(4) + P(6) = 3/6$$

Probability as Area

What is the probability of

- Selecting a yellow ball?
- $E = \{Y1, Y2, \dots, Y25\}$

$$\bullet \quad P(E) = \frac{\# E}{\# S} = \frac{25}{30+25+25} = \frac{5}{16}$$

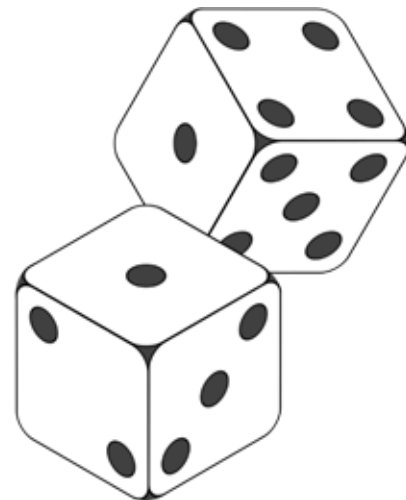


$$P(\text{Yellow}) = P(Y1) + \dots + P(Y25) = 25 * (1/80)$$

- Suppose we throw two fair dice
 - What is the sample space S (space of all possible outcomes)?

Event E : the two dice's outcomes sum to 6

- What is the size of E ?
- What is the probability of E ?

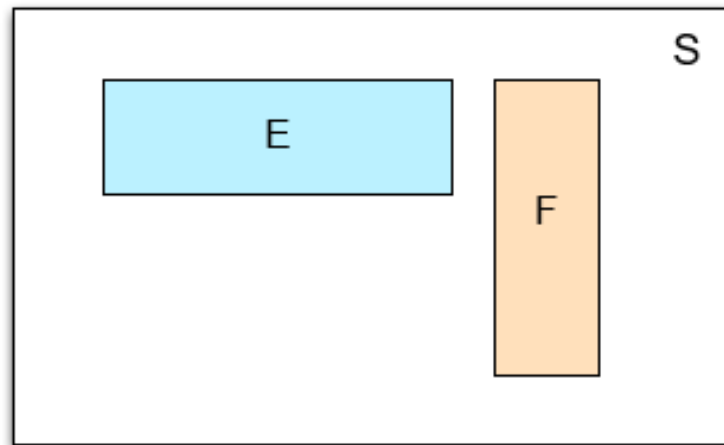


Disjoint events

- In general, breaking an event into *disjoint events* preserves the total probability
- **Disjoint:** E and F are *disjoint* if they cannot happen simultaneously,
- e.g. $E = \{2, 4\}$, $F = \{1, 3, 5\}$

- In such cases,

$$P(E \text{ or } F) = P(E) + P(F)$$



Elementary events

- Notice that we can find the total probability of an event by breaking it into pieces and adding up the probabilities of the pieces:

$$P(\text{Even}) = P(2 \text{ or } 4 \text{ or } 6) = P(2) + P(4) + P(6) = 3/6$$

1	2 $P(2) = 1/6$	3
4 $P(4) = 1/6$	5	6 $P(6) = 1/6$

- These pieces are called ‘**elementary events**’:
Events that correspond to exactly one outcome
- These elementary events are **disjoint events**

Partition

- We say that disjoint events $E_1, \dots, E_n$ form a *partition* of E if any outcome in E lies in exactly one E_i
- e.g.
 - {Fr.} E_1 , {Soph.} E_2 form a partition of {Lower division} E
 - {Fr.}, {Soph.} {J.} {Sen.} form a partition of S (sample space)

Freshmen	Sophomores	Juniors	Seniors
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Probability as Area

For disjoint events E, F :

$$P(E \text{ or } F) = P(E) + P(F)$$

If disjoint events $E_1, \dots, E_n$ forms a partition of E :

$$P(E) = P(E_1) + P(E_2) + \dots + P(E_n)$$

If disjoint events $E_1, \dots, E_n$ forms a partition of S , for event F (law of total probability):

$$P(F) = P(E_1, F) + P(E_2, F) + \dots + P(E_n, F)$$

Notation: $P(A, B)$ is a shorthand for $P(A \text{ and } B)$

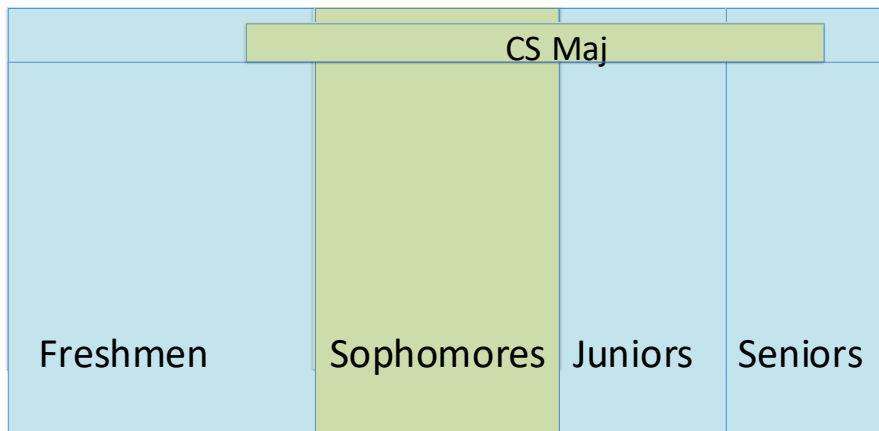
Probability as Area

Examples

- $P(\text{CS}) = P(\text{Fr.}, \text{CS}) + P(\text{Soph.}, \text{CS}) + P(\text{J.}, \text{CS}) + P(\text{Sen.}, \text{CS})$

Notation: $P(A, B)$ is a shorthand for $P(A \text{ and } B)$

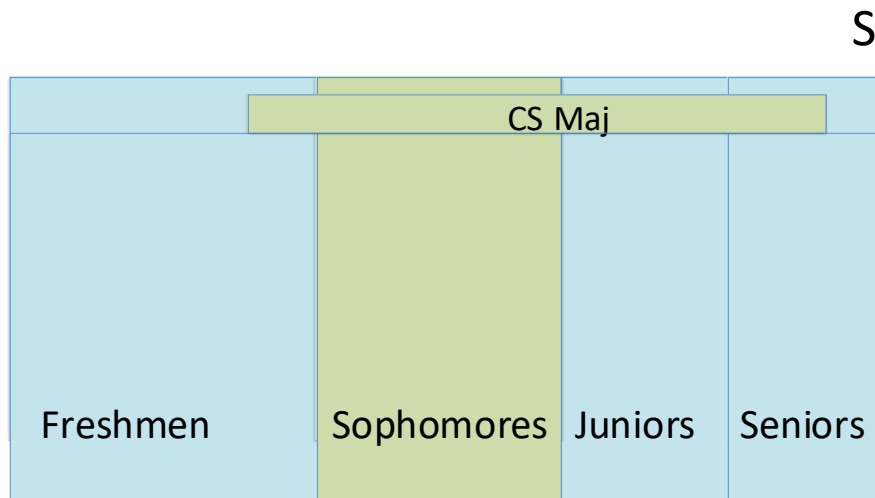
- $P(\text{Soph.}) = P(\text{CS}, \text{Soph.}) + P(\text{nonCS}, \text{Soph.})$
S



Probability as Area

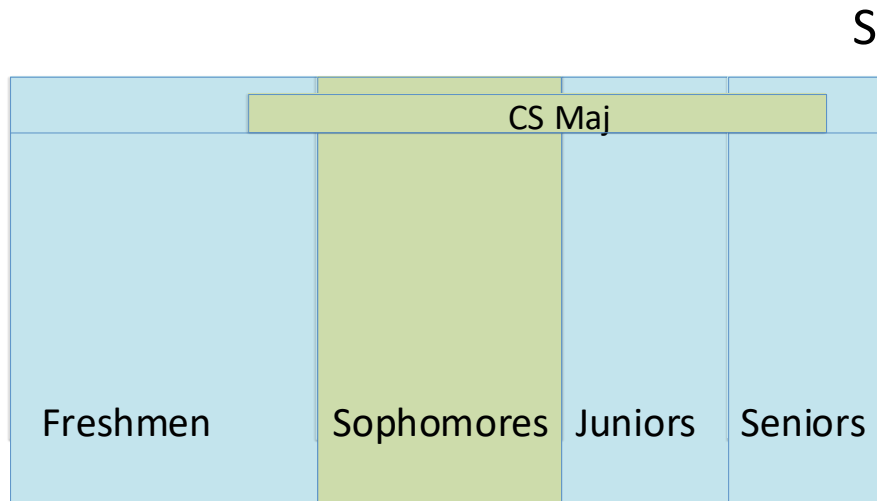
If events E, F are non-disjoint, what is $P(E \text{ or } F)$?

- $P(\text{Soph. or CS})$?
- Note: events “Sophomore” and “CS Major” may overlap



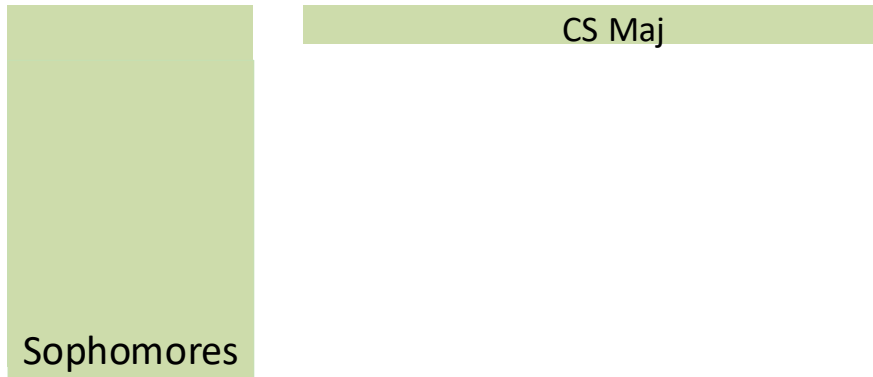
Probability as Area

- $E = \{ \text{Soph OR CS} \}$
- Is $P(E) = P(\text{Soph}) + P(\text{CS})$?
 - No
- Which one is larger?
 - Let's see..



Probability as Area

$$P(\text{Soph}) + P(\text{CS}) =$$



$$P(\text{Soph or CS}) =$$

